

Donghyun Son

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EDUCATION

Stanford University, <i>M.S. in Computer Science</i>	Sep 2026 — Jun 2028 (expected)
Seoul National University, <i>B.S. in Computer Science</i>	Mar 2018 — Feb 2026
GPA: 4.04/4.3 (4.11/4.3 in major), <i>Summa Cum Laude</i> , includes 3 years of mandatory military service	
University of Texas at Austin, <i>Exchange Student, Computer Science</i>	Sep 2025 — Dec 2025

PUBLICATIONS

*: equal contribution.

Preprints

- [1] *Tailoring the Quantization Space for 1-Bit KV Cache Compression*
Minsoo Cheong*, **Donghyun Son***, Sungjoo Yoo
- [2] *NALAR: A Serving Framework for Agent Workflows*
Benedict Marco Laju, **Donghyun Son**, Saurabh Agarwal, Nitin Kedia, Myungjin Lee, Jayanth Srinivasa, Aditya Akella

Conference & Workshop Papers

- [1] *EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models*
Minsoo Cheong, **Donghyun Son**, Woosang Lim, Sungjoo Yoo
NeurIPS 2026, AdaptFM@ICML 2026
- [2] *NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache*
Donghyun Son, Euntae Choi, Sungjoo Yoo
NeurIPS 2025
- [3] *Gradient Estimation for Unseen Domain Risk Minimization with Pre-Trained Models*
Byounggyu Lew*, **Donghyun Son***, Buru Chang
OOD-CV@ICCV 2023
- [4] *Reliable Decision from Multiple Subtasks through Threshold Optimization: Content Moderation in the Wild*
Donghyun Son*, Byounggyu Lew*, Kwanghee Choi*, Yongsu Baek, Seungwoo Choi, Beomjun Shin, Sungjoo Ha, Buru Chang
WSDM 2023 (Oral)

Technical Reports

- [1] *K-EXAONE 2.0 Technical Report*
LG AI Research, contributed to inference infrastructure

WORK EXPERIENCE

AI Scientist Intern, EXAONE Lab <i>LG AI Research</i>	Apr 2026 — Jul 2026 <i>Seoul, South Korea</i>
- Integrated and optimized EXAONE models across LLM serving frameworks (vLLM, SGLang) - Implemented EXAONE model support for Ollama, including MTP-based speculative decoding - Investigated reinforcement learning pipelines to identify sources of training–inference mismatch	
Undergraduate Researcher <i>UTNS Research Group @ UT Austin, Supervised by Prof. Aditya Akella</i>	Aug 2025 — Dec 2025 <i>Austin, Texas</i>
Designed and implemented NALAR, a serving framework that dynamically schedules, migrates, and scales stateful agent workflows using workflow-level visibility and two-level control. (under review)	
Undergraduate Researcher <i>CMALab @ SNU, Supervised by Prof. Sungjoo Yoo</i>	Sep 2024 — Aug 2025 / Jul 2026 — Sep 2026 <i>Seoul, South Korea</i>
- Investigated a calibration-free vector quantization method for KV cache, incorporating a novel normalization algorithm and efficient CUDA kernels for 1–2 bit inference (NeurIPS25) - Proposed a vector quantization method for KV cache which tailors the quantization space to reflect error sensitivity and inter-channel dependencies (under review)	

Machine Learning Engineer, Moderation ML Team <i>Hyperconnect (Acquired by Match Group)</i>	Mar 2022 — Jul 2023 <i>Seoul, South Korea</i>
• Built an ML-based system for moderating images and audio across Match Group brands (e.g., Tinder, Hinge) • Devised a threshold optimization algorithm based on gradient descent to optimize the tradeoff between reliability and cost (WSDM23, Oral) • Proposed a training algorithm for ML models to generalize effectively across different services and domains (OOD-CV@ICCV23)	
Research Engineer, R&D Team <i>VisualCamp</i>	Aug 2020 — Mar 2022 <i>Seoul, South Korea</i>
• Designed and implemented C++-based multi-threaded ML inference pipeline for on-device low-latency gaze estimation • Built a training pipeline robust to data corruption and reduced MSE of a gaze estimation model by 30%	

HONORS & AWARDS

Competitions

- **Master**, at Codeforces (handle: **diordhd**, rating: 2200+)
- **2nd prize**, at Union of Clubs Programming Contest (UCPC) 2025
- **5th prize**, at Samsung Collegiate Programming Contest (SCPC) 2021
- **5th place**, at ICPC NERC Huawei Challenge 2020: Cloud Scheduling Challenge
- **Bronze Prize**, at 42nd Undergraduate Mathematics Contest Division 1 (for mathematical majors)

Scholarships

- **Study Abroad Scholarship of Kwanjeong Educational Foundation** (\$60,000), from Kwanjeong Educational Foundation
- **Samsung Software Membership**, from Samsung Research (Nov 2021 — Feb 2026)
- **Korea-U.S. STEM Student Exchange Scholarship** (\$9,000), from Ministry of Trade, Industry and Energy (2025.09)
- **Merit-based Full Tuition Academic Scholarship** (\$2,300), from Seoul National University (2020-1, 2023-2, 2024-1, 2024-2)

PRESENTATIONS

- **KV Cache Compression for Long Context Inference**, Weekly Seminar @ Deepest S16 (January 2025)
- **Efficient Algorithms for LLM Inference**, Weekly Seminar @ Deepest S15 (July 2024)

ADDITIONAL INFORMATION

Technical Skills: C++, Python, Pytorch, Tensorflow, Jax, CUDA, OpenCL, MPI, OpenMP, vLLM, SGLang
Conference & Workshop Reviewer: NeurIPS 2026, AdaptFM@ICML 2026, LCFM@NeurIPS 2026